

Relationship Types & Relationship Sets

Q. Explain the concepts of Relationship Types and Relationship Sets. Discuss the Degree of Relationship and Mapping Cardinalities (1:1, 1:N, M:N) with examples.

📌 Definition to Remember

A **Relationship Type** defines a meaningful association between two or more entity types, represented by a **Diamond** in ER diagrams. A **Relationship Set** is the actual collection of specific relationship instances existing in the database at any given moment.

1. Relationship Type and Relationship Set

- **Relationship Type:** Association between entity types. E.g., **WORKS_FOR** links **EMPLOYEE** and **DEPARTMENT**.
- **Relationship Set:** All specific instances of that relationship. E.g., "John works for IT Dept." is one instance.

2. Degree of Relationship

The **Degree** = number of entity types participating in the relationship.

Degree	Name	Example
1	Unary	Employee Manages another Employee
2	Binary	Student Enrolls_In Course (most common)
3	Ternary	Doctor, Patient, Drug connected by Prescribes

3. Mapping Cardinalities

Mapping Cardinality defines the maximum number of relationship instances an entity can participate in.

1. One-to-One (1:1)

- One entity in A associates with **at most one** in B, and vice versa.
- **Example:** One **MANAGER** manages one **DEPARTMENT**; one **DEPARTMENT** has one **MANAGER**.

2. One-to-Many (1:N)

- One entity in A associates with **multiple** entities in B; but each B associates with **one** A.
- **Example:** One **DEPARTMENT** has many **EMPLOYEE**s; each **EMPLOYEE** belongs to one **DEPARTMENT**.

3. Many-to-One (N:1)

- Multiple entities in A associate with **one** entity in B.
- **Example:** Many **STUDENT**s belong to one **COURSE**.

4. Many-to-Many (M:N)

- Entities in A can associate with **multiple** in B, and vice versa.
- Example:** A **STUDENT** enrolls in many **COURSE** s; a **COURSE** has many **STUDENT** s.

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1:1 → A — <R> — B
1:N → A == <R> — B
M:N → A == <R> == B
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4. Participation Constraints

Type	Description	ER Symbol
Total Participation	Every entity in the set MUST participate	Double Line (==)
Partial Participation	Entity may or may not participate	Single Line (—)

- Total Example:** Every **EMPLOYEE** must belong to a **DEPARTMENT** .
- Partial Example:** A **STUDENT** may or may not live in a **HOSTEL** .

★ Must-Write Points (for 10 marks)

- Relationship Type = association between entity types (Diamond); Relationship Set = collection of all instances.
- Degree: Unary (1), Binary (2 — most common), Ternary (3).
- Mapping Cardinalities: 1:1, 1:N, N:1, M:N — define how many entities can participate.
- 1:1: one entity maps to exactly one (e.g., Manager → Department).
- 1:N: one on one side, many on the other (e.g., Department → Employees).
- M:N: many on both sides — requires a Junction Table in the relational model.
- Total Participation (double line): every entity **MUST** participate; Partial (single line): optional.

⚡ Quick Recall

Relationship Type (Diamond) → Degree (Unary/Binary/Ternary) → Cardinality (1:1, 1:N, M:N) → Participation (Total=double line, Partial=single line)